

5-4

Solving Radical Equations

I CAN... solve radical equations and inequalities.

VOCABULARY

- extraneous solution

LOOK FOR RELATIONSHIPS

When you square both sides of an equation, you are multiplying each side by the same quantity. The expression for the quantity differs, but $\sqrt{x+5}$ and 4 are equal.

EXPLORE & REASON

- Solve $3(a+1)^2 + 2 = 11$. Use at least two different methods.
- Try each of the methods you used in part (a) to solve $3\sqrt{a+1} + 2 = 11$.
- Generalize** Which of the methods is better suited for solving an equation with a radical? What problems arise when using the other method?

ESSENTIAL QUESTION

What properties of exponents allow you to solve equations that include radicals or rational exponents?

EXAMPLE 1 Solve an Equation With One Radical

- Solve the radical equation $\sqrt{x+5} - 1 = 3$.

To solve this equation, you can isolate the radical. Then you can square both sides of the equation to eliminate the radical and solve for x .

$$\sqrt{x+5} - 1 = 3$$

$$\sqrt{x+5} = 4$$

$$(\sqrt{x+5})^2 = 4^2$$

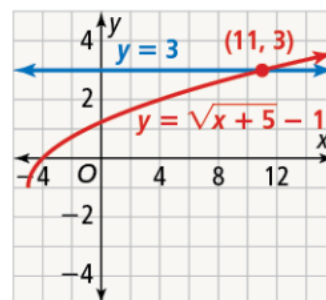
$$x+5 = 16$$

$$x = 11$$

Square both sides to eliminate the radical.

So the solution to the radical equation is $x = 11$.

You can check by graphing $y = \sqrt{x+5} - 1$ and $y = 3$ on the same coordinate axes. The graphs of the equations intersect at $(11, 3)$.



- Solve the radical equation $\sqrt[3]{x} + 2 = 4$.

$$\sqrt[3]{x} + 2 = 4$$

$$\sqrt[3]{x} = 2$$

$$(\sqrt[3]{x})^3 = 2^3$$

$$x = 8$$

Cube both sides to eliminate the cube root.

So the solution to the radical equation is $x = 8$.

Check your answer by substituting 8 for x in the original equation:

$$\sqrt[3]{8} + 2 = 2 + 2 = 4$$

Try It! 1. Solve each radical equation. Check your solution by substituting into the given equation or graphing.

a. $\sqrt{x-2} + 3 = 5$

b. $\sqrt[3]{x-1} = 2$

EXAMPLE 2 Rewrite a Formula

The suspension cables from the Golden Gate Bridge's towers are farther above the roadway near the towers and closer to the roadway near the middle of the bridge. You can figure out your distance from the middle of the bridge, x , in feet, and the height of the suspension cable, y , in feet, by using the given equation. About how far is the cable from the roadway when you are 200 ft from the middle of the bridge?



Rewrite the equation to isolate y so you can express the vertical distance between the roadway of the bridge and the suspension cable in terms of x .

$$x = \frac{\sqrt{y - 220}}{0.010583}$$

$$0.010583x = \sqrt{y - 220}$$

$$(0.010583x)^2 = (\sqrt{y - 220})^2$$

$$0.000112x^2 \approx y - 220$$

$$0.000112x^2 + 220 \approx y$$

$$0.000112(200)^2 + 220 \approx y$$

$$224.48 \approx y$$

This equation gives the vertical distance between the deck and the suspension cable for different values of x .

COMMON ERROR

When squaring a term that includes a coefficient and a variable, remember to square both parts of the term. If you have to round the coefficient after squaring, be sure to use the approximately equal sign.

The cable is about 224 ft above the bridge's roadway when you are 200 ft from the middle of the bridge.

Try It! 2. The speed, v , of a vehicle in relation to its stopping distance, d , is represented by the equation $v = 3.57\sqrt{d}$. What is the equation for the stopping distance in terms of the vehicle's speed?

EXAMPLE 3 Identify an Extraneous Solution

A. Solve the radical equation $\sqrt{3x - 2} = x - 4$.

$$\sqrt{3x - 2} = x - 4$$

$$(\sqrt{3x - 2})^2 = (x - 4)^2$$

$$3x - 2 = x^2 - 8x + 16$$

$$0 = x^2 - 11x + 18$$

$$0 = (x - 9)(x - 2)$$

$$x = 9 \text{ or } x = 2$$

Check the potential solutions by substituting the solutions for x in the original equation:

$$\sqrt{3x - 2} = x - 4$$

$$\sqrt{3x - 2} = x - 4$$

$$\sqrt{3(9) - 2} = 9 - 4$$

$$\sqrt{3(2) - 2} = 2 - 4$$

$$\sqrt{25} = 5 \checkmark$$

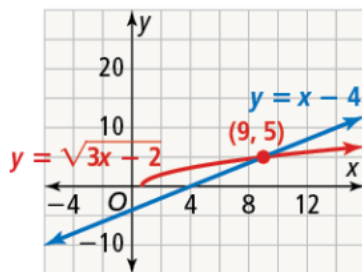
$$\sqrt{4} \neq -2 \times$$

So 9 is the only solution to the equation and 2 is an extraneous solution.

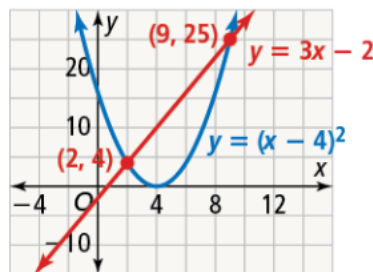
B. Why does this extraneous solution arise?

The first step in solving the equation in part (a) was to square both sides.

Graph on one grid, $y = \sqrt{3x - 2}$ and $y = x - 4$. Then graph on another grid $y = 3x - 2$ and $y = (x - 4)^2$.



These intersect at (9, 5), so the solution to $\sqrt{3x - 2} = x - 4$ is $x = 9$.



These intersect at (2, 4) and (9, 25), so the solutions to $(\sqrt{3x - 2})^2 = (x - 4)^2$ are $x = 2$ and $x = 9$.

This means that the equations $\sqrt{3x - 2} = x - 4$ and $(\sqrt{3x - 2})^2 = (x - 4)^2$ are not equivalent equations. Both graphs have an intersection point at $x = 9$, but the second graph also has an intersection point at $x = 2$. Squaring both sides of the original equation created an extraneous solution.

VOCABULARY

An **extraneous solution** is a solution of an equation *derived* from an equation, but it is *not* a solution of the original equation.

Try It! 3. Solve each radical equation. Check for any extraneous solutions.

a. $x = \sqrt{7x + 8}$

b. $x + 2 = \sqrt{x + 2}$

EXAMPLE 4 Solve Equations With Rational Exponents**A.** What are the solutions to the equation $(x^2 + 5x + 5)^{\frac{5}{2}} = 1$?

$$(x^2 + 5x + 5)^{\frac{5}{2}} = 1$$

$$\left((x^2 + 5x + 5)^{\frac{5}{2}}\right)^{\frac{2}{5}} = (1)^{\frac{2}{5}}$$

Raise both sides to the reciprocal power.

$$x^2 + 5x + 5 = 1$$

$$x^2 + 5x + 4 = 0$$

$$(x + 4)(x + 1) = 0$$

$$x + 4 = 0 \text{ or } x + 1 = 0$$

Use the Zero-Product Property.

$$x = -4 \text{ or } x = -1$$

Check for extraneous solutions.

$$((-4)^2 + 5(-4) + 5)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$((-1)^2 + 5(-1) + 5)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$(16 - 20 + 5)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$(1 - 5 + 5)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$(1)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$(1)^{\frac{5}{2}} \stackrel{?}{=} 1$$

$$1 = 1 \checkmark$$

$$1 = 1 \checkmark$$

This equation has two solutions, $x = -4$ and $x = -1$. There are no extraneous solutions.**B.** What is the solution to $(x + 18)^{\frac{3}{2}} = (x - 2)^3$?

$$(x + 18)^{\frac{3}{2}} = (x - 2)^3$$

$$\left((x + 18)^{\frac{3}{2}}\right)^{\frac{2}{3}} = \left((x - 2)^3\right)^{\frac{2}{3}}$$

Raise both sides to the reciprocal power.

$$x + 18 = (x - 2)^2$$

$$x + 18 = x^2 - 4x + 4$$

$$x^2 - 5x - 14 = 0$$

$$(x + 2)(x - 7) = 0$$

$$x + 2 = 0 \quad x - 7 = 0$$

$$x = -2 \quad x = 7$$

Use the Zero-Product Property.

Check for extraneous solutions.

$$(-2 + 18)^{\frac{3}{2}} \stackrel{?}{=} (-2 - 2)^3$$

$$(7 + 18)^{\frac{3}{2}} \stackrel{?}{=} (7 - 2)^3$$

$$16^{\frac{3}{2}} \stackrel{?}{=} (-4)^3$$

$$25^{\frac{3}{2}} \stackrel{?}{=} 5^3$$

$$64 \neq -64 \times$$

$$125 = 125 \checkmark$$

So 7 is the only solution to the equation, and -2 is an extraneous solution.**STUDY TIP**

An equation can have one solution, multiple solutions, or no solutions, so it's important to check all potential solutions.

Try It! 4. Solve each equation.

a. $(x^2 - 3x - 6)^{\frac{3}{2}} - 14 = -6$

b. $(x - 8)^{\frac{3}{2}} = (10 - x)^3$

EXAMPLE 5 Solve an Equation With Two RadicalsSolve the radical equation $\sqrt{x+9} - \sqrt{2x} = 3$.

When an equation has two radicals, start the solution process by isolating one radical.

COMMON ERROR

A common mistake when squaring an expression like $\sqrt{2x} + 3$ is to only square the radical portion. Square this expression as you would any binomial, by using the Distributive Property.

$$\sqrt{x+9} - \sqrt{2x} = 3$$

$$\sqrt{x+9} = \sqrt{2x} + 3$$

Isolate one radical sign.

$$(\sqrt{x+9})^2 = (\sqrt{2x} + 3)^2$$

$$x + 9 = 2x + 6\sqrt{2x} + 9$$

$$0 = x + 6\sqrt{2x}$$

$$-x = 6\sqrt{2x}$$

Isolate the remaining radical and square again.

$$(-x)^2 = (6\sqrt{2x})^2$$

$$x^2 - 72x = 0$$

$$x(x - 72) = 0$$

$$x = 0 \text{ or } x = 72$$

Check the potential solutions by substituting 0 and 72 for x in the original equation:

$$\sqrt{x+9} - \sqrt{2x} \stackrel{?}{=} 3$$

$$\sqrt{x+9} - \sqrt{2x} \stackrel{?}{=} 3$$

$$\sqrt{(0)+9} - \sqrt{2(0)} \stackrel{?}{=} 3$$

$$\sqrt{(72)+9} - \sqrt{2(72)} \stackrel{?}{=} 3$$

$$\sqrt{9} - \sqrt{0} \stackrel{?}{=} 3$$

$$\sqrt{81} - \sqrt{144} \stackrel{?}{=} 3$$

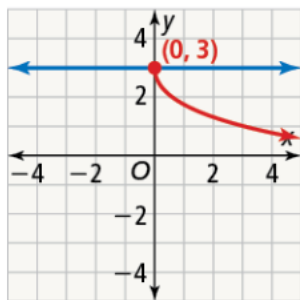
$$3 - 0 \stackrel{?}{=} 3$$

$$9 - 12 \stackrel{?}{=} 3$$

$$3 = 3 \quad \checkmark$$

$$-3 \neq 3 \quad \times$$

The only solution is 0. The value 72 does not make the original equation true, so it is an extraneous solution.

You can also see this by graphing $y = \sqrt{x+9} - \sqrt{2x}$ and $y = 3$ together. They intersect only at $x = 0$.**Try It!** 5. Solve each radical equation. Check for extraneous solutions.

a. $\sqrt{x+4} - \sqrt{3x} = -2$

b. $\sqrt{15-x} - \sqrt{6x} = -3$

APPLICATION

EXAMPLE 6 Solve a Radical Inequality

The body surface area (BSA) of a human being is used to determine some doses of medication. The formula for finding BSA is $BSA = \sqrt{\frac{H \cdot M}{3,600}}$, where H is the height in centimeters and M is the mass in kilograms.

A doctor calculates a particular dose of medicine for a patient. What must the mass of the person be for the dose to be appropriate?



Substitute in an expression equal to BSA.

$$\begin{aligned} BSA &< 1.9 \\ \sqrt{\frac{H \cdot M}{3,600}} &< 1.9 \\ \sqrt{\frac{160 \cdot M}{3,600}} &< 1.9 \\ \left(\sqrt{\frac{160 \cdot M}{3,600}}\right)^2 &< 1.9^2 \\ \frac{160 \cdot M}{3,600} &< 3.61 \\ M &< 81.225 \end{aligned}$$

Substitute 160 for H , and solve for M .

CONSTRUCT ARGUMENTS

Another way to solve this problem is to substitute values into the formula, solve it as an equation, and then use logic to determine the inequality symbol. When there are many possible solution methods, how do you decide which to use?

The mass of the patient must be less than 81.225 kg for the dose to be appropriate.

Try It! 6. A doctor calculates that a particular dose of medicine is appropriate for an individual whose BSA is less than 1.8. If the mass of the individual is 75 kg, how many cm tall can he or she be for the dose to be appropriate?

CONCEPT SUMMARY Solving Radical Equations

	WORDS	ALGEBRA	GRAPH
Step 1	Isolate the radical term.	$2\sqrt{x+3} - x = 0$ $2\sqrt{x+3} = x$	
Step 2	Square both sides to remove the radical.	$(2\sqrt{x+3})^2 = (x)^2$	
Step 3	Solve the equation.	$4(x+3) = x^2$ $x^2 - 4x - 12 = 0$ $(x-6)(x+2) = 0$ $x = 6 \text{ or } x = -2$	
Step 4	Eliminate extraneous solutions.	$2\sqrt{6+3} - 6 \stackrel{?}{=} 0$ $2\sqrt{9} \stackrel{?}{=} 6$ $6 \stackrel{?}{=} 6$ $6 = 6 \checkmark$	$2\sqrt{-2+3} - (-2) \stackrel{?}{=} 0$ $2\sqrt{1} \stackrel{?}{=} -2$ $2 \stackrel{?}{=} -2$ $2 \neq -2 \times$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** What properties of exponents allow you to solve equations that include radicals or rational exponents?
- Construct Arguments** How can you use a graph to show that the solution to $\sqrt[3]{84x+8} = 8$ is 6?
- Vocabulary** Why does solving a radical equation sometimes result in an extraneous solution?
- Error Analysis** Nazim said that -3 and 6 are the solutions to $\sqrt{3x+18} = x$. What error did Nazim make?
- Communicate Precisely** Describe how you would solve the equation $x^{\frac{2}{3}} = n$. How is this solution method to be interpreted if the equation had been written in radical form instead?

Do You KNOW HOW?

Solve for x .

6. $3\sqrt{x+22} = 21$

7. $\sqrt[3]{5x} = 25$

In 8 and 9, find the extraneous solution.

8. $\sqrt{8x+9} = x$

9. $x = \sqrt{24-2x}$

10. Rewrite the equation $y = \sqrt{\frac{x-48}{6}}$ to isolate x .

11. Use a graph to find the solution to the equation $9 = \sqrt{3x+11}$.

Solve each equation.

12. $(3x+2)^{\frac{2}{3}} = 4$

13. $\sqrt{2x-5} - \sqrt{x-3} = 1$

14. $\sqrt{x+2} + \sqrt{3x+4} = 2$

UNDERSTAND

15. **Generalize** Explain how to identify an extraneous solution for an equation containing a radical expression.
16. **Look for Relationships** Write a radical equation that relates a square's perimeter to its area. Explain your reasoning. Use s to represent the side length of the square.
17. **Error Analysis** Describe and correct the error a student made in rewriting the equation to isolate y .

$$x = \frac{\sqrt{58 + y}}{1.98}$$

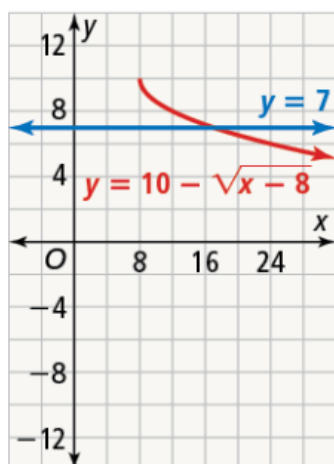
$$1.98x = \sqrt{58 + y}$$

$$1.98x^2 = 58 + y$$

$$1.98x^2 - 58 = y$$



18. **Use Appropriate Tools** Find the point of intersection of the two graphs.



19. **Higher Order Thinking** Some, but not all, equations with rational exponents have extraneous solutions. What is the relationship between the exponents and the possibility of having extraneous solutions for equations with rational exponents? Explain your reasoning.
20. **Communicate Precisely** Describe the process used to solve an equation with two radical expressions. How is this process different from solving an equation with only one radical expression?

PRACTICE

Solve each radical equation. SEE EXAMPLE 1

21. $\sqrt[3]{x} + 8 = 13$
22. $\sqrt{4x} = 11$
23. $\sqrt{75 + x} - 6 = 14$
24. $25 - \sqrt[4]{x} = 22$

Solve for y . SEE EXAMPLE 2

25. $x = 3(\sqrt[3]{15 + y})$
26. $x = \frac{\sqrt{2y}}{26}$
27. $x = \frac{\sqrt{y - 14.2}}{0.05}$
28. $x = \frac{1}{3}(\sqrt[4]{y})$

Solve each radical equation. Check for extraneous solutions. SEE EXAMPLE 3

29. $x = \sqrt{x + 6}$
30. $2x = \sqrt{17x - 15}$
31. $4x = \sqrt{6x + 10}$
32. $x = \sqrt{56 - x}$

Solve. SEE EXAMPLE 4

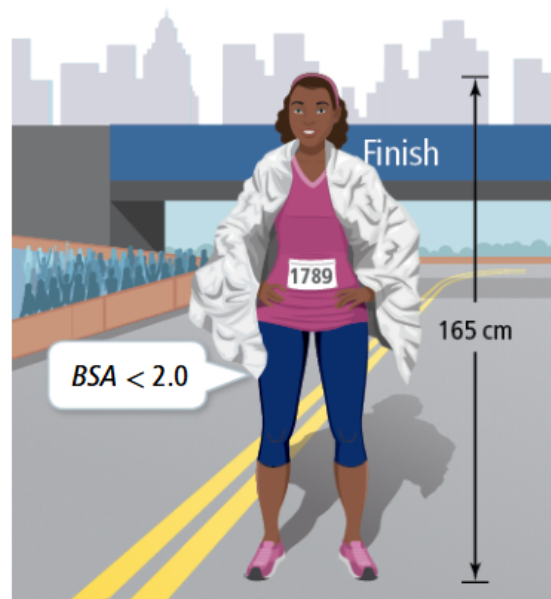
33. $0.5(x^2 + 5x + 136)^{\frac{2}{3}} = 50$
34. $2(x^2 - 12x - 4)^{\frac{1}{2}} - 3 = 15$
35. $(x^2 + 4x + 5)^{\frac{3}{2}} + 1 = 0$

Solve each radical equation. Check for extraneous solutions. SEE EXAMPLE 5

36. $\sqrt{6 + x} - \sqrt{x - 5} = 2$
37. $\sqrt{4x + 5} - \sqrt{x + 1} = 1$
38. $\sqrt{x + 1} + 1 = \sqrt{x + 3}$

Solve using the formula $BSA = \sqrt{\frac{H \cdot M}{3,600}}$. SEE EXAMPLE 6

39. A sports medicine specialist determines that a hot-weather training strategy is appropriate for this athlete. To the nearest tenth, what can the mass of the athlete be for the training strategy to be appropriate?



PRACTICE & PROBLEM SOLVING

APPLY

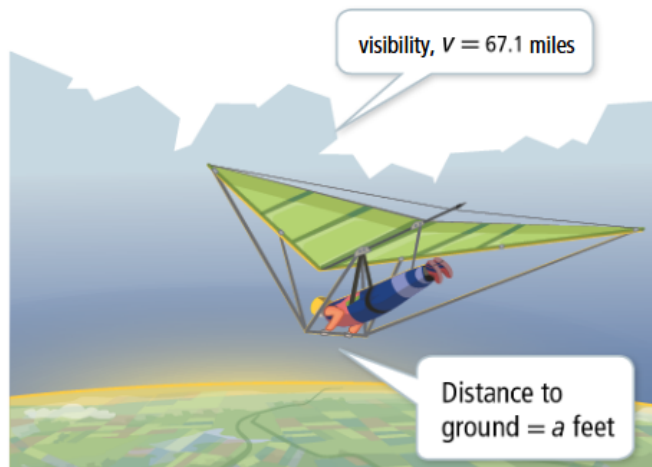
40. Specialists can determine the speed a vehicle was traveling from the length of its skid marks, d , and the coefficient of friction, f . Rewrite the formula to solve for the length of the skid marks.



41. **Make Sense and Persevere** The half-life of a certain type of soft drink is 5 h. If you drink 50 mL of this drink, the formula $y = 50(0.5)^{\frac{t}{5}}$ tells the amount of the drink left in your system after t hours. How much of the soft drink will be left in your system after 16 hours?

42. **Model With Mathematics** Big Ben's pendulum takes 4 s to swing back and forth. The formula $t = 2\pi\sqrt{\frac{L}{32}}$ gives the swing time, t , in seconds, based on the length of the pendulum, L , in feet. What is the minimum length necessary to build a clock with a pendulum that takes longer than Big Ben's pendulum to swing back and forth?

43. **Make Sense and Persevere** Derek is hang gliding on a clear day. Use the formula for visibility, $v = 1.225\sqrt{a}$ to find the altitude at which Derek is hang gliding.



ASSESSMENT PRACTICE

44. Complete the table to solve for the unknown value in the equation $y = \sqrt[3]{2x + z} - 12$, using the given values in each row.

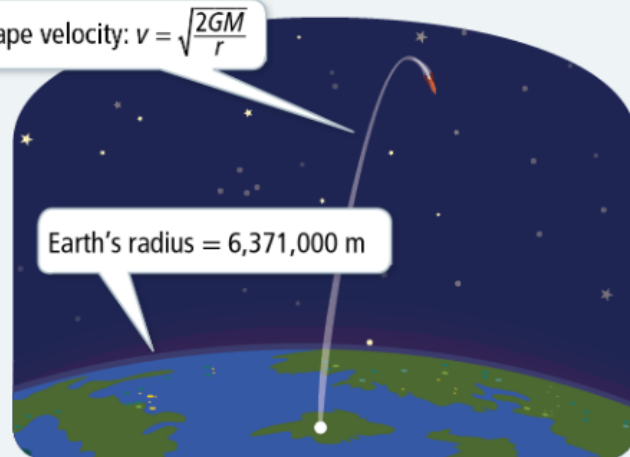
y	x	z
0	462	
-3		439
-10	1.25	
3		16

45. **SAT/ACT** What is the solution to the equation $(x^2 + 5x + 25)^{\frac{3}{2}} = 343$?

- (A) -8 only
- (B) 3 only
- (C) 77 only
- (D) -8 and 3
- (E) There are no solutions.

46. **Performance Task** Escape velocity is the velocity at which an object must be traveling to leave a star or planet without falling back to its surface or into orbit. Escape velocity depends on the gravitational constant, $G = 6.67 \times 10^{-11}$, the mass, M in kilograms, and radius, r , of the star or planet.

Escape velocity: $v = \sqrt{\frac{2GM}{r}}$



Part A Rewrite the equation to solve for mass.

Part B Earth has an escape velocity of about 11,200 meters per second. What is Earth's mass in kilograms?



The Snack Shack

Americans seem to love the beach! When the weather is warm, people flock to the beach. Some people bring coolers packed with food and drinks. Others prefer to take advantage of snack bars and shops set up along the beach.

Some beachside communities have built long wooden walkways, or boardwalks, to make it easier for beachgoers to walk to the snack bars and stores. How easy do you find walking in the sand? Think about this during the Mathematical Modeling in 3 Acts lesson.

ACT 1 Identify the Problem

1. What is the first question that comes to mind after watching the video?
2. Write down the main question you will answer about what you saw in the video.
3. Make an initial conjecture that answers this main question.
4. Explain how you arrived at your conjecture.
5. What information will be useful to know to answer the main question? How can you get it? How will you use that information?

ACT 2 Develop a Model

6. Use the math that you have learned in this Topic to refine your conjecture.

ACT 3 Interpret the Results

7. Did your refined conjecture match the actual answer exactly? If not, what might explain the difference?